Do Cross-Asset and Media Features Help Predict Cryptocurrency Crash Regimes?

An Empirical Ablation Study on BTC and ETH

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Abstract

We investigate whether adding external feature sources — media-derived signals from GDELT and cross-asset market indicators from the paired cryptocurrency — meaningfully improves the detection of crash regimes in Bitcoin (BTC) and Ethereum (ETH). The baseline for each asset is a gradient boosting model trained on 17 standard market features derived from daily OHLCV data. We then run eight additional configurations that layer media features, cross-asset features, or both. The target label identifies days at which the closing price forms a local high relative to a 7-day sliding window (five past days plus the next day), which corresponds to crash-onset periods and, to a lesser extent, recovery peaks. Across all nine experiments and six model families, the answer to our central question is consistently negative: no external feature source improves on the market-only baseline for either asset. For BTC the best PR-AUC is 0.771 (market only); for ETH it is 0.180. Every enriched configuration performs strictly worse. We discuss why this result is not altogether surprising and what it implies for future work.

Keywords: Bitcoin, Ethereum, crash detection, cross-asset features, media signals, imbalanced classification, gradient boosting, ablation study.

1. Introduction

The goal of this project was simple: take a working baseline model for BTC and ETH crash detection and check whether two cheap external signal sources — news media coverage and the price dynamics of the paired asset — could push the performance up. The question is practically motivated. If ETH price behavior contains information about an upcoming BTC crash, or if a spike in media attention precedes a downturn, that would be useful and easy to exploit operationally.

We did not start from a blank slate. The baseline models already incorporate 17 engineered market features per asset (returns, volatility, drawdown, volume indicators), selected and validated through exploratory analysis combining Spearman correlation and mutual information. The external features are added incrementally on top of this baseline, making the ablation clean: any change in performance can be attributed to the added signal family.

The short answer is that neither source helps. This paper documents the methodology, the exact feature sets tested, and the results in enough detail to understand why, and to motivate better-targeted alternatives.

2. Related Work

Several studies have explored the prediction of Bitcoin price movements using machine learning, mostly framed as direction prediction (up/down) rather than crash detection. Gradient boosting models tend to outperform linear classifiers and neural networks on tabular financial data when training sets are of moderate size. Sentiment derived from Twitter, Reddit, and Google Trends has shown modest positive effects on return predictability in some periods but is highly unstable across market regimes. Cross-asset effects between BTC and ETH are well-documented in the volatility spillover literature but less studied in the context of discrete event prediction.

The evaluation of rare-event classifiers through PR-AUC rather than accuracy or ROC-AUC is now standard in imbalanced learning. For time-series applications, preserving chronological order in train–test splits is necessary to avoid look-ahead bias.

3. Data

3.1. Market Data

Daily OHLCV data for BTCUSDT and ETHUSDT are from the Binance public API. BTC covers 3,115 days from 2017-09-16 to 2026-03-27; ETH covers 3,114 days over the same period through 2026-03-26. The raw pipeline follows a Bronze (ingestion) → Silver (cleaning, type casting, deduplication) → Gold (feature engineering + label) architecture.

3.2. Media Data

Media features are derived from the GDELT Project, a publicly available database of worldwide media events. After keyword filtering and daily aggregation, the gold table contains 4,035 daily observations (2015-02-18 to 2026-03-24), with two main signals: `article_count` (daily article volume) and `avg_tone` (mean sentiment score). Rolling averages and lags of these two variables are computed as supplementary features.

3.3. Merging

Market and media datasets are joined by date using an inner merge. Cross-asset configurations merge BTC and ETH market datasets first (yielding 3,114 rows), then optionally add media. All merges use inner joins, so row counts are governed by the shorter series.

4. Target Definition

The binary label is constructed as follows. For each day t, the implementation computes

$$M_t = \texttt{close.shift(-1).rolling(7).min()} = \min(P_{t-5},\, P_{t-4},\, \ldots,\, P_t,\, P_{t+1})\,, \tag{1}$$

and then sets $y_t = 1$ if $M_t/P_t - 1 < -0.10$.

This fires whenever today's closing price exceeds by more than 10% at least one price in the window $\{P_{t-5}, \ldots, P_{t+1}\}$. The window is **mostly retrospective**: it looks back five days and only one day forward. In practice this means the label identifies days when the market is at a local cyclical high relative to its recent neighborhood — which typically corresponds to crash-onset days (the price is about to fall) but also to recovery days following a crash (the price has bounced back above a recent low still within the 5-day lookback).

This is an important nuance: the task is not pure forward crash prediction. Features like `drawdown` or `return_7d` are structurally correlated with this label, which partly explains the high PR-AUC values for BTC. A strictly forward-looking label ($M_t = \min_{1 \le k \le 7} P_{t+k}$) would be more demanding and is the natural next step.

Positive-class rates: $p_{\mathrm{BTC}} = 12.2\%$ (approx. 380 days out of 3,115), $p_{\mathrm{ETH}} = 3.7\%$ (approx. 115 days out of 3,114).

5. Features

5.1. Market Features (Baseline, 17 per Asset)

All features are computed from information available at time t. Redundant features are dropped using a pairwise Pearson correlation threshold of 0.9.

Table 1: Market feature set (17 features, same structure for BTC and ETH).

Category	Features
Returns & lags	`return_1d`, `return_7d`, `lag_return_1d`, `lag_return_7d`
Volatility & lag	`volatility_7d`, `volatility_30d`, `lag_volatility_7d`
Drawdown & structure	`drawdown`, `ma_ratio`, `momentum_acc`, `momentum_volatility`
Volume & buy pressure	`volume`, `volume_norm`, `lag_volume_norm`, `quote_asset_volume`, `number_of_trades`, `buy_pressure`, `lag_buy_pressure`

Dropped for multicollinearity: raw OHLC (`open`, `high`, `low`), simple moving averages `ma_7`/`ma_30`, `taker_buy_base_volume`, and `pressure_x_return` for BTC ($\rho = 0.998$ with `return_1d`).

5.2. Media Features

Media features are selected using a combined score $s(j) = |\rho_j^{\mathrm{Sp}}| / \max |\rho^{\mathrm{Sp}}| + \mathrm{MI}_j / \max \mathrm{MI}$, where normalization is over all features jointly. Features with $s(j) < 0.10$ are dropped.

For **BTC**, 7 features are retained: three article-count derivatives (`article_count_ma_3`, `article_count`, `article_count_lag_1`) and four tone derivatives (`tone_ma_30`, `tone_ma_7`, `tone_lag_7`, `tone_lag_1`). The raw `avg_tone` is dropped ($s = 0.10$, $\mathrm{MI} = 0$). Top score: `article_count_ma_3` at $s = 0.82$.

For **ETH**, only 3 features pass the threshold: `avg_tone` ($s = 0.17$, $\mathrm{MI} = 0$), `article_count_ma_3` ($s = 0.13$), `tone_ma_7` ($s = 0.13$). The maximum MI of any ETH media feature (≈ 0.004) is nine times lower than for BTC (≈ 0.036), indicating that GDELT carries essentially no information for ETH crash detection.

5.3. Cross-Asset Features

For BTC-target experiments, ETH market columns are renamed with a `_eth` suffix and merged by date. Raw OHLCV and non-stationary columns (`open`, `high`, `low`, `close`, `ma_7`, `ma_30`) are excluded, leaving 17 engineered ETH features. A reduced set of 5 top-scoring ETH features is also tested (05b).

For ETH-target experiments, BTC features are processed symmetrically; 14 of the 28 BTC candidate features pass the combined-score threshold of 0.10.

6. Experimental Setup

Train/test split. Chronological 80/20 split: roughly 2,492 training days and 623 test days for BTC.

Hyperparameter optimization. 100 Optuna trials per model (TPE sampler, MedianPruner), optimizing global out-of-fold PR-AUC over a five-fold TimeSeriesSplit on the training set. The test set is never used during optimization.

Models. Six families: DummyClassifier (majority class), Logistic Regression (`class_weight=balanced`), Random Forest (`class_weight=balanced`), XGBoost, LightGBM, CatBoost.

Evaluation. Primary metric: PR-AUC on the test set. Secondary: Precision, Recall, F1 at a threshold tuned on OOF predictions to maximize F1.

Configurations. Nine total, structured as an ablation over three signal families (Table 2).

Table 2: Feature configurations. "Market" = 17 engineered OHLCV features of the target asset.

ID	Configuration	Target	Market	Cross	Media	Total
01	Market only	BTC	17	–	–	17
03	Market + Media	BTC	17	–	7	24
05	Market + Cross (17)	BTC	17	17	–	34
05b	Market + Cross (top-5)	BTC	17	5	–	22
08	Market + Cross + Media	BTC	17	17	7	41
02	Market only	ETH	17	–	–	17
04	Market + Media	ETH	17	–	3	20
06	Market + Cross (14)	ETH	17	14	–	31
07	Market + Cross + Media	ETH	17	14	3	34

7. Results

7.1. BTC

Table 3 summarizes the best PR-AUC per configuration. The market-only model (01, CatBoost) is the best across all five BTC configurations. Adding media features (03) costs −0.011 in PR-AUC; adding 17 ETH cross-features (05) costs −0.006; reducing to 5 ETH features (05b) gives the same result as using 17; and combining all sources (08) is the worst at −0.017.

Table 3: BTC results — best-model PR-AUC per configuration. Dummy PR-AUC = 0.149.

ID	Best model	PR-AUC	Precision	Recall	F1
01	**CatBoost**	**0.771**	0.745	0.704	0.724
05	CatBoost	0.766	0.704	0.704	0.704
05b	CatBoost	0.766	0.716	0.727	0.722
03	CatBoost	0.760	0.711	0.700	0.705
08	CatBoost	0.755	0.660	0.751	0.702
Dummy	–	0.149	0.158	0.779	0.263

The full model breakdown for notebook 01 is shown in Table 4. CatBoost, XGBoost, and LightGBM are closely grouped; logistic regression lags by 0.065 points, confirming that a non-trivial share of the signal is non-linear.

Table 4: All models on notebook 01 (BTC, 17 market features).

Model	PR-AUC	Precision	Recall	F1
CatBoost	**0.771**	0.745	0.704	0.724
XGBoost	0.762	0.722	0.708	0.715
LightGBM	0.752	0.748	0.680	0.712
Random Forest	0.744	0.653	0.735	0.692
Logistic Regression	0.706	0.490	0.877	0.629
Dummy	0.149	0.158	0.779	0.263

7.2. ETH

Results for ETH are shown in Table 5. The overall level is much lower: the best configuration (02, market only, LightGBM) achieves PR-AUC = 0.180, barely above the dummy at 0.035. The same pattern holds: every external source degrades performance, with the combined configuration (07) dropping to 0.152.

Table 5: ETH results — best-model PR-AUC per configuration. Dummy PR-AUC = 0.035.

ID	Best model	PR-AUC	Precision	Recall	F1
02	**LightGBM**	**0.180**	0.182	0.544	0.273
06	CatBoost	0.179	0.203	0.353	0.258
04	LightGBM	0.169	0.195	0.441	0.270
07	CatBoost	0.152	0.160	0.647	0.257
Dummy	–	0.035	0.036	0.441	0.067

Table 6: All models on notebook 02 (ETH, 17 market features).

Model	PR-AUC	Precision	Recall	F1
LightGBM	**0.180**	0.182	0.544	0.273
XGBoost	0.173	0.177	0.632	0.277
CatBoost	0.167	0.171	0.544	0.260
Random Forest	0.160	0.161	0.676	0.261
Logistic Regression	0.101	0.098	0.735	0.173
Dummy	0.035	0.036	0.441	0.067

8. Discussion and Critique

8.1. Why the External Features Did Not Help

The most plausible explanation is signal dilution. Gradient boosting with 100 Optuna trials explores the hyperparameter space reasonably well at 17 features, but as the feature count grows to 34 or 41, the same budget is spread over a higher-dimensional optimization landscape, and the model is more likely to partially overfit on low-signal features. This interpretation is supported by the pattern of degradation: adding five ETH cross-features (05b) and seventeen (05) produce the same PR-AUC for BTC (0.766), meaning the twelve additional features in 05 contribute zero information but add noise.

Logistic regression is more sensitive to this effect. Its BTC PR-AUC drops from 0.706 (notebook 01) to 0.517 when seven media features are added (notebook 03) — a 0.189 deterioration — because linear models cannot adaptively down-weight uninformative features.

The GDELT media signal for ETH is essentially non-existent: the maximum mutual information of any ETH media feature is ≈ 0.004, against ≈ 0.036 for BTC. This alone would prevent any media-driven improvement on ETH. For BTC the signal is real but modest, and insufficient to offset the dilution cost.

8.2. Critique: Was This the Right Question?

Looking back, the cross-feature experiment had a structural weakness. The ETH features added to the BTC model are engineered from the same type of data (OHLCV), using the same feature construction pipeline. In a market where BTC and ETH are highly correlated (daily returns routinely show $\rho > 0.7$ over extended periods), the ETH cross-features carry information that largely overlaps with what BTC's own market features already encode. Testing 17 or even 5 ETH features on top of 17 BTC features is therefore unlikely to introduce genuinely orthogonal information.

A more targeted design would have isolated periods of BTC/ETH decorrelation and checked whether ETH features help specifically during those windows. Alternatively, using ETH-specific structural variables — DeFi protocol flows, staking ratios, L2 activity — that have no direct BTC counterpart would provide a cleaner test of cross-asset incremental value.

The same critique applies to GDELT. Mainstream news media follows markets; it does not typically lead them, especially for intraday and single-day crash events. Testing media features that come from social platforms closer to the trader community (Twitter, Reddit, Telegram) would be a more credible hypothesis.

8.3. The Target Definition Issue

The label as implemented (Section 4) is mostly retrospective: it detects local price highs within a window of five past days plus one future day. This means the model is partly learning to identify *days already inside* a crash episode, not exclusively days that *precede* one. Features like `drawdown` and `return_7d` are almost definitionally correlated with this label, which inflates the apparent PR-AUC, particularly for BTC. The strong results for BTC should therefore be interpreted cautiously: some of the signal may not generalize to truly out-of-sample future crash prediction. A forward-only label would be a more honest baseline for any follow-up study.

9. Conclusion

We set out to answer a simple question: does adding GDELT media features or ETH/BTC cross-asset features improve crash-regime detection? The answer, across nine configurations and six model families, is no.

For BTC, the market-only CatBoost model achieves PR-AUC = 0.771, and every enriched configuration performs worse. For ETH, the task is much harder to begin with (only 3.7% positive rate), and the best model barely clears 0.180 regardless of the feature set used.

Three factors explain the failure of external features to add value: (i) GDELT media signals for ETH carry near-zero mutual information with the target; (ii) ETH market features are highly correlated with BTC market features, making cross-asset additions largely redundant; and (iii) a fixed Optuna budget of 100 trials is increasingly inefficient as the feature count grows.

More broadly, the experiment exposes a design weakness that is common in exploratory ML studies: adding correlated data sources from the same distributional family (OHLCV of a correlated asset) rarely produces orthogonal signal. Future work should focus on genuinely independent information sources — on-chain data, funding rates, platform-native sentiment — and should correct the target definition to be strictly forward-looking before drawing conclusions about true predictive power.

Data and Code Availability

All data processing scripts and benchmark notebooks are available in the project repository. Market data is sourced from the Binance public API; GDELT data is freely available at `https://www.gdeltproject.org`.